Particle Physics 1 Basic Concepts

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Chapter 1

New Revolutions in Particle Physics: Basic Concepts

These notes follow Lecture 1 of Leonard Susskind's *Particle Physics 1: Basic Concepts*. The story is told, as in the lecture, for logical force rather than strict chronology, and the curation of the lecture materials is due to LazyingArt LLC. The opening question is ancient and still governs the subject: is nature discrete, or is it continuous? From that question we are led, step by step, through chemistry, radioactivity, light as a classical wave, light as a quantum, relativity, natural units, and finally to the practical rule that drives particle physics: to see smaller things we need shorter wavelengths, and shorter wavelengths require larger momentum and energy.

1.1 Discreteness, Continuity, and the First Chemical Evidence

Particle physics is introduced here in the broadest possible way. We do not begin with a catalogue of particles. We begin with two rival pictures of matter.

The first picture says that matter comes in indivisible units. The ancient word was *atom*; in the lecture Susskind immediately relaxes the language and says: let us just call such discrete units particles. The second picture says the opposite: matter is not built from localized lumps at all, but is continuously distributed, smeared out through space. In that language one speaks of fields, and the lecture is careful to say what a field is: a function on space. A density is a field. The electric field is a field. The magnetic field is a field. A field is not a little object sitting somewhere; it is something defined throughout a region of space.

Already at the beginning we are warned not to expect a simple victory for one side. Quantum mechanics will eventually force us to say that the particle picture and the continuum picture are both right and both incomplete. But before that subtlety appears, we want evidence.

The first serious evidence for discreteness enters through chemistry. Dalton noticed that the mass of a fixed number of molecules of a substance comes, to good approximation, in integer multiples of the mass of the same number of hydrogen molecules. The point is not exact arithmetic perfection. The point is that the ratios are near integers. Not three halves, not π, but integers. That is just the sort of pattern one expects if matter is being built from some underlying units.

From the modern point of view we can see what Dalton was really glimpsing. Ordinary matter is

made from atoms, and the mass of an atom is concentrated overwhelmingly in protons and neutrons. Electrons are very light by comparison and contribute only a small correction. Protons and neutrons have nearly equal mass, and the nucleus of hydrogen contains just one proton. Therefore the mass of ordinary matter is, to first approximation, counted in units of proton-or-neutron mass, and Dalton's integral pattern becomes intelligible. The lecture does not overstate this: it is an approximation, not an exact law, but it is a very revealing approximation.

By the end of nineteenth-century chemistry the elements themselves looked like the elementary building blocks. There were roughly a hundred kinds of atoms, and it was natural to think that the story stopped there. Then, as the lecture puts it, all hell broke loose.

1.2 Radioactivity and the First Particle Beam

The next step is radioactivity, and the lecture treats it as the first real particle-physics experiment. The story begins with a happy accident. A radioactive source affects a photographic plate as if some invisible radiation had struck it. Becquerel then tries to study that radiation more carefully. He buries the source in a block of lead, drills a narrow hole through the lead, and places a photographic plate some distance away. The hole collimates the emission, and what had been a vague effect becomes a narrow beam.

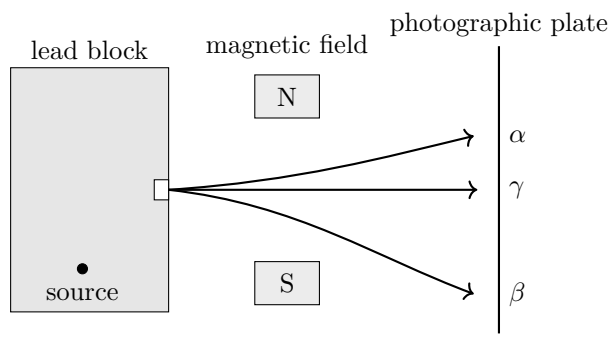

Figure 1.1: A schematic version of Becquerel's collimated radioactive source. In the lecture this becomes the first particle beam, and a magnetic field separates the radiation into α, β, and γ components.

That is already a striking development: something is being emitted in a directed beam. Then a magnetic field is brought near the beam, and the beam separates into three branches. One branch goes straight through, as if the magnetic field were absent. The other two bend in opposite directions, as charged objects should in a magnetic field, and they do not bend equally strongly.

From the modern point of view the identifications are

$$\beta\text{-radiation} = \text{electrons}, \qquad \alpha\text{-radiation} = \text{helium nuclei}, \qquad \gamma\text{-radiation} = \text{photons}. \tag{1.1}$$

But the lecture does not move too quickly. Before telling us what these branches are, it pauses over what was actually learned. Whatever was coming out of the source behaved as if it came in distinguishable components, and if the source were dilute enough one could imagine the screen registering arrivals one at a time: blip, blip, blip. Alpha arrives in little bundles. Beta arrives in little bundles. Gamma too arrives in little bundles.

At this stage γ-radiation is the real mystery. It is neutral, it reaches the screen one event at a time, and yet there is no familiar neutral particle available to explain it. The lecture uses exactly that unresolved tension to pivot into the next subject. Gamma, it turns out, is light. But once gamma is recognized as light, a deeper problem appears, because light already seemed to belong to the opposite camp.

1.3 Light as a Classical Wave

Gamma radiation is announced to be light, or rather very high-energy light. But that statement does not immediately settle anything. It sharpens the problem. For light was the strong classical example of something continuous, something field-like, something that looked very much not like a particle.

The lecture first secures the classical case before permitting any quantum rescue. Suppose electromagnetic radiation is emitted from a source. The source might be a macroscopic antenna; it might equally well be an atom, which in this context acts as a perfectly good microscopic antenna. If light were merely a spray of bullets, then a small hole in an obstruction should produce a small sharp spot on a screen. Instead one gets a fuzzier region. That is suggestive, but not decisive. One might still imagine particles being deflected by the edges of the opening.

The decisive experiment uses two openings rather than one. If light were only particles, then two holes would simply give two smeared spots superimposed on one another. But that is not what happens. One sees alternating bright and dark bands. The bright bands come from constructive interference, where wave crests meet wave crests. The dark bands come from destructive interference, where a crest from one opening meets a trough from the other.

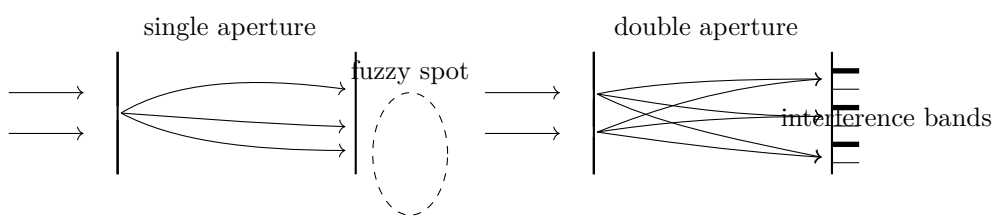

Figure 1.2: Single-aperture spreading and double-aperture interference. In the lecture the first effect is suggestive, while the second really nails the wave character of light.

That is the experiment that really nails the wave theory. Maxwell then tells us what kind of wave light is. It is a wave of electric and magnetic field. If the wave propagates along some axis, the electric field oscillates transversely, the magnetic field oscillates transversely as well, and the two are perpendicular to one another.

Once that classical picture is secure, the lecture pauses and deliberately slows down. Now we introduce notation. This is not a digression. It is preparation. The next equations will matter because photons will inherit them.

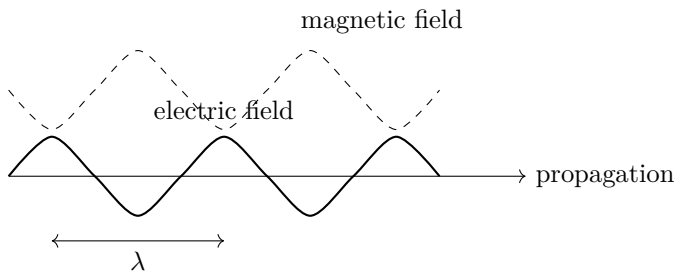

Figure 1.3: A schematic electromagnetic wave. The lecture first secures light as a classical wave of electric and magnetic field before allowing the quantum puzzle to appear.

1.4 Wave Kinematics

The first quantity introduced is the wavelength. It is simply the spatial length of one full cycle of the wave. The board evidence for that moment is worth keeping, because the lecture begins a fresh little derivation there.

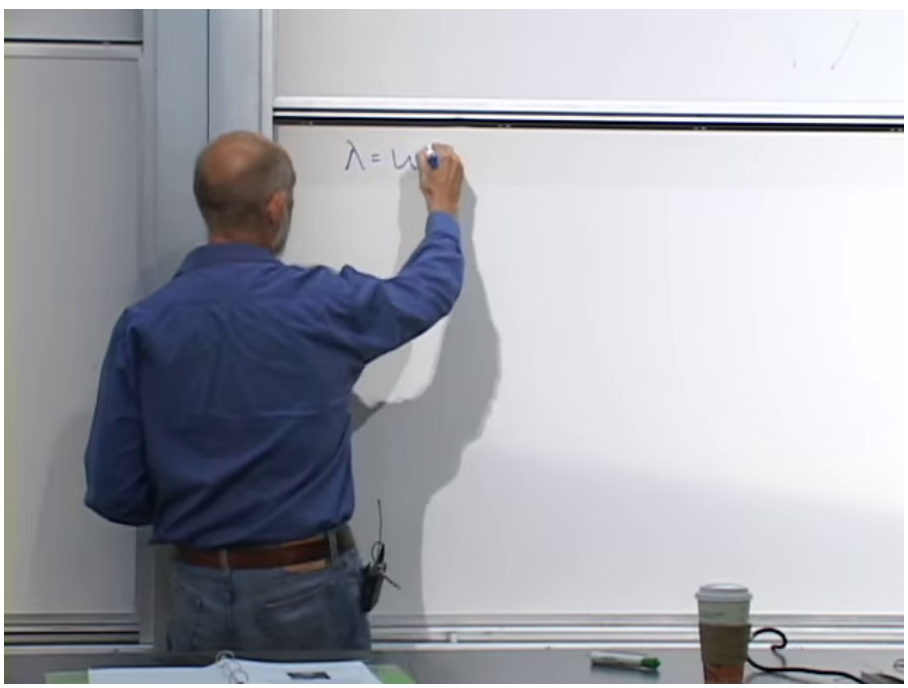

Figure 1.4: Introducing λ as wavelength. The board writing is only partially visible, but the transcript makes the intended notation clear.

So we write

$$\lambda = \text{wavelength}. \tag{1.2}$$

Next comes the temporal description. If we stand still and let the wave pass us, the time required for one full oscillation is the period, which we denote by T. In one period the wave advances by one wavelength. Therefore its speed is distance divided by time:

$$v = \frac{\lambda}{T}. \tag{1.3}$$

For light the speed is c, so

$$c = \frac{\lambda}{T}. \tag{1.4}$$

It is then convenient to replace the period by its inverse, the frequency:

$$f = \frac{1}{T}. \tag{1.5}$$

Substituting this into the previous equation gives the relation written on the board,

$$\lambda f = c. \tag{1.6}$$

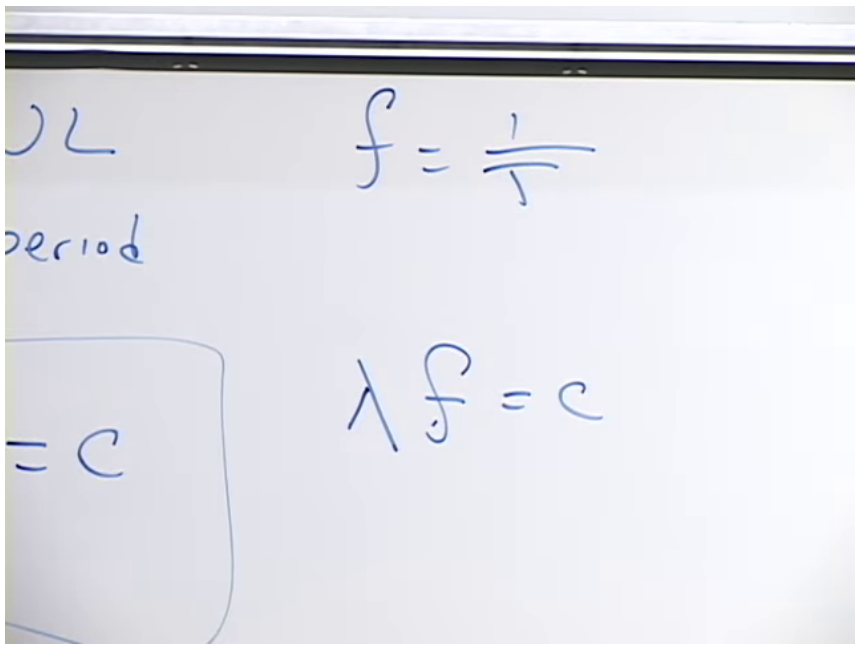

Figure 1.5: Frequency and wavelength relations on the board. The lecture has now moved from period to frequency and then to the light-wave relation $\lambda f = c$.

That still uses cycles per second. The lecture now shifts to the notation physicists prefer. Instead of counting full turns per second, we count radians per second and define the angular frequency

$$\omega = 2\pi f. \tag{1.7}$$

The board also shows the equivalent rearrangement $\omega/(2\pi) = f$. The mathematical reason for preferring ω is modest but real: once one writes many wave formulas, the factors of 2π are usually better controlled in terms of angular frequency than ordinary frequency.

Combining $\omega = 2\pi f$ with $f = c/\lambda$ gives the key relation:

$$\omega = \frac{2\pi c}{\lambda}. \tag{1.8}$$

Equivalently,

$$\omega\lambda = 2\pi c. \tag{1.9}$$

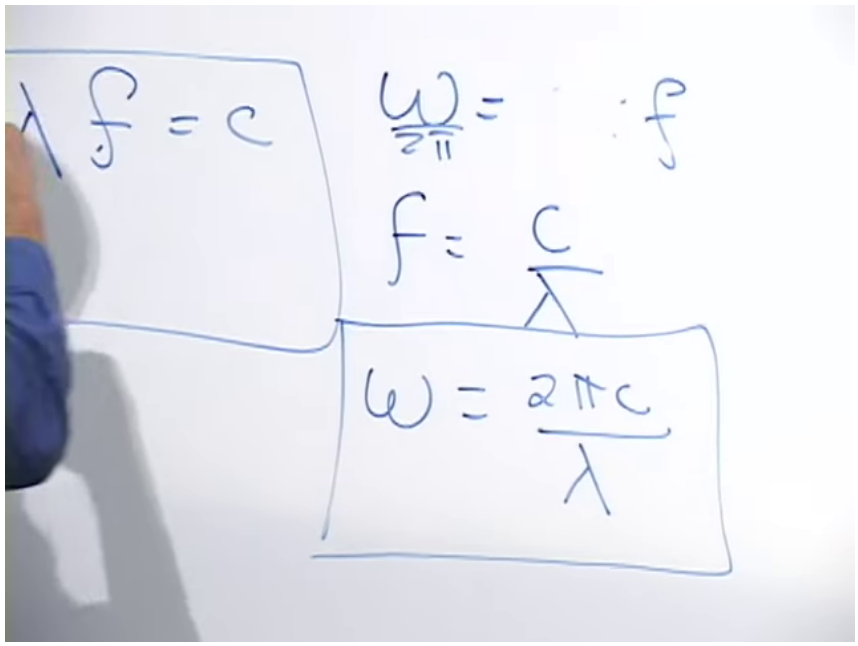

Figure 1.6: The board sequence culminating in the boxed relation $\omega = \frac{2\pi c}{\lambda}$. The boxed placement makes clear that this is the takeaway equation of the mini-derivation.

Let us keep the whole chain together:

$$\begin{aligned} c &= \frac{\lambda}{T}, \\ f &= \frac{1}{T}, \\ \lambda f &= c, \\ f &= \frac{c}{\lambda}, \\ \omega &= 2\pi f, \\ \omega &= \frac{2\pi c}{\lambda}. \end{aligned} \tag{1.10}$$

The lecture has now done exactly what it needed to do. It has built a small vocabulary of waves in the order required for the next transition. Once λ, f, and ω are live variables, the quantum puzzle can be stated sharply.

1.5 Photons, Probabilities, and Energy

We are now ready for the conceptual obstacle around which the lecture turns.

1.5.1 Question & Answer

Question. If light is a wave, why does it arrive as isolated blips?

Answer. The lecture asks us to imagine attenuating a beam of light more and more. At ordinary intensity the screen shows a continuous blob, or, in a two-hole experiment, a continuous interference pattern. But if the beam is weakened enough, the character of what we see changes. The smooth illumination gives way to isolated events, little dots, little blips.

If we wait long enough, however, those separate arrivals build up the same pattern the classical wave would have produced. That is the crucial point. The arrivals are discrete, but the distribution of where they arrive is governed by the wave pattern. So the classical wave theory survives, not as a literal continuous smearing of substance, but as a rule for probabilities. The wave tells us how likely a photon is to appear in one place rather than another.

This is the first quantum-mechanical lesson of the lecture. The photon arrives as an indivisible event, but the pattern of arrivals is wave-like. And the lecture immediately broadens the point: electrons behave the same way. Send them one at a time through two holes, and they build up an interference pattern. Heavier objects do too. The lecture mentions alpha particles, then buckyballs, and even points toward the possibility of still larger objects. So the mysterious union of wave and particle is not a peculiarity of light. It is a basic feature of the quantum world.

Once that point is made, the lecture pivots to a quantitative question. What does a photon carry? The answer is energy, and the energy of one photon is

$$E_\gamma = \hbar\omega. \tag{1.11}$$

This equation now has real content because ω has already been related to wavelength. Using (1.10), shorter wavelength means larger ω, and therefore larger photon energy:

$$\lambda \downarrow \qquad \Longrightarrow \qquad \omega \uparrow \qquad \Longrightarrow \qquad E_\gamma \uparrow . \tag{1.12}$$

The lecture then emphasizes the specifically quantum remnant of discreteness. A light beam of fixed frequency does not carry an arbitrary continuum of energy. If it contains n photons, each of frequency ω, then

$$E_{\text{beam}} = n\,\hbar\omega, \qquad n = 0, 1, 2, \ldots \tag{1.13}$$

Thus a radio beam may carry a lot of total energy, but only because it contains many photons. Each individual radio photon has low energy. A gamma-ray photon, by contrast, has very short wavelength and therefore high energy even one at a time.

This is where the lecture cashes out the quantum lesson in practical terms. If we want to see smaller structures, we need shorter wavelengths. But shorter wavelengths mean higher frequencies, and higher frequencies mean larger photon energies. So the road to smaller scales already leads us toward energetic probes.

1.6 Relativity, Natural Units, and the Return to Momentum

At this point the lecture explicitly says that it is not abandoning the previous line of thought. It is adding another one. Particle physics is relativistic quantum mechanics, so the next indispensable ingredient is relativity.

The famous equation is

$$E_{\text{rest}} = mc^2. \tag{1.14}$$

But the lecture immediately cleans up the language. In modern usage, *mass* means what older textbooks called *rest mass*. We do not say that the mass of an electron increases when it moves. The energy changes with motion; the mass does not. So the equation above refers to the energy of a massive object in a frame in which the object as a whole is at rest.

The lecture gives two examples to make that concrete.

First, take a box of gas resting on a scale. Heat it. The molecules move more violently, so internal energy has been added, even though the box remains at rest as a whole. The point is that the rest mass of the box increases along with that internal energy. In differential form one may summarize the point as

$$\Delta E_{\text{rest}} = \Delta m\, c^2. \tag{1.15}$$

The effect is tiny in ordinary situations, but the meaning is unmistakable: for a body at rest, mass and energy are the same physical content written in different units.

Second, take an electron and a positron at rest and let them annihilate. Their rest energy must reappear as photon energy. The reaction is schematically

$$e^- + e^+ \to \gamma + \gamma, \tag{1.16}$$

and the total energy released is

$$E_{\text{annihilation}} = 2m_e c^2. \tag{1.17}$$

The lecture even pauses over a student's question about mechanism. We may not yet know the detailed mechanism, but conservation of energy already forces the conclusion that if the pair disappears, its energy must show up elsewhere.

The discussion then turns to units, and here the lecture becomes more philosophical. We have three basic kinds of units in ordinary mechanics: length, time, and mass. Because units are choices, one can use that freedom to simplify equations by setting certain conversion constants equal to one. In particle physics the two constants most naturally chosen are c and $\hbar$:

$$c = 1, \qquad \hbar = 1. \tag{1.18}$$

Why these two? The lecture's answer is that they are universal. The speed of light is a universal limiting speed; no object outruns it. Planck's constant is universal because quantum mechanics constrains every object by the same scale. The lecture phrases this in terms of the uncertainty principle: the product of position-uncertainty and momentum-uncertainty is bounded below by a number of order $\hbar$. One may speak cautiously here and write

$$\Delta x\, \Delta p \gtrsim \hbar, \tag{1.19}$$

not as a new derivation in this chapter, but as the lecture's reason for calling $\hbar$ universal.

There is also a third universal constant, G, and if one sets $c = \hbar = G = 1$ one is led to Planck units. But the lecture is clear: ordinary particle physics is essentially insensitive to gravity, so it is not helpful here to normalize G. That choice belongs more naturally to quantum gravity, cosmology, or black-hole physics.

With $c = 1$, the rest-energy relation becomes especially simple.

1.6.1 Question & Answer

Question. What does $E = m$ mean, and why is that not the same statement as the photon-energy equation?

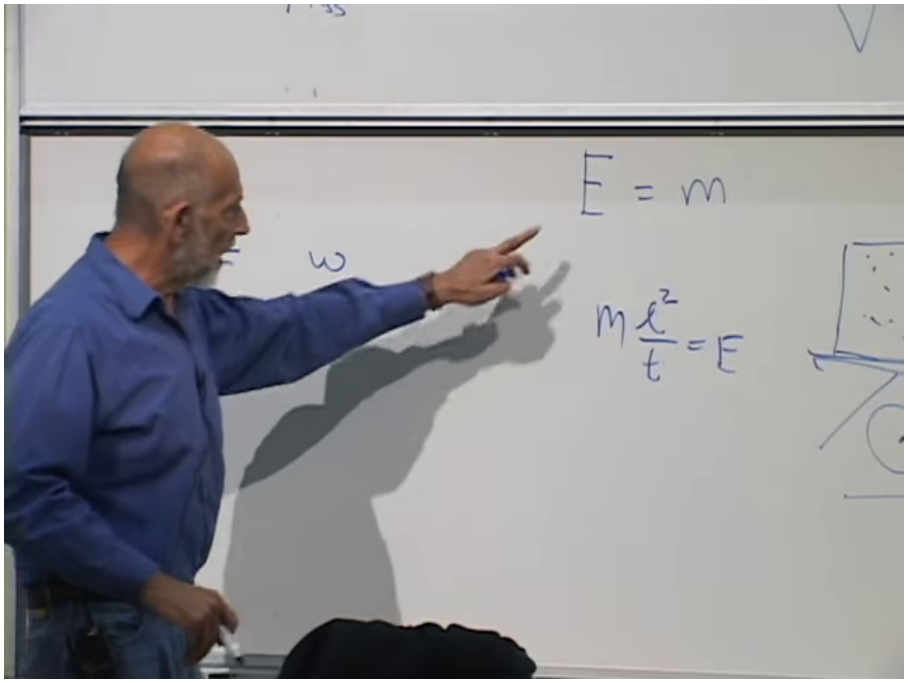

Figure 1.7: Board evidence for the natural-units form $E = m$. The visible statement is the simplification of rest energy after setting $c = 1$; the lower handwritten line is too uncertain to use here as an independent equation.

Answer. Once we choose units with $c = 1$, the rest-energy equation becomes

$$E_{\text{rest}} = m. \tag{1.20}$$

This is a simplification of the relation for a massive object in its rest frame. It does not mean that every form of energy is literally a mass, and it certainly does not mean that photons have rest mass.

For photons the relevant equation is different. A photon cannot be brought to rest. Its energy is not rest energy. If we also choose units with $\hbar = 1$, then

$$E_\gamma = \omega. \tag{1.21}$$

The two formulas look parallel because natural units remove conversion constants from both of them, but their meanings are different. One is the rest energy of a massive particle. The other is the energy of a massless quantum of light.

The lecture then returns to momentum, because momentum is the quantity that finally ties the wave formulas and the photon formulas back to the practical logic of experiments. For slow-moving bodies,

$$\mathbf{p} = m\mathbf{v}. \tag{1.22}$$

But that is not the appropriate formula for light. The lecture simply states the correct relation for a collimated light beam:

$$p = \frac{E}{c}, \tag{1.23}$$

where p is the magnitude of the beam momentum. This already explains a qualitative fact. Light clearly carries energy; it heats things. Its momentum is harder to notice because it is energy divided

by the very large number c. Ordinary light therefore gives only a tiny push unless the beam is intense.

Now the lecture does one final blackboard exercise. Take a single photon. Its momentum is its energy divided by c, but its energy is $\hbar\omega$. Therefore

$$\begin{aligned} p_\gamma &= \frac{E_\gamma}{c} \\ &= \frac{\hbar\omega}{c}. \end{aligned} \tag{1.24}$$

Next use the wave relation $\omega = 2\pi c/\lambda$. Substituting,

$$\begin{aligned} p_\gamma &= \frac{\hbar}{c}\left(\frac{2\pi c}{\lambda}\right) \\ &= \frac{2\pi\hbar}{\lambda} \\ &= \frac{h}{\lambda}. \end{aligned} \tag{1.25}$$

That is the final closure. The wavelength is not merely a geometrical label for a wave. It is directly tied to momentum. Shorter wavelength means larger momentum. And since photon energy is also proportional to frequency, hence inversely related to wavelength, shorter wavelength means larger energy as well.

So we arrive at the central chain of the lecture:

$$\text{smaller structure} \implies \text{shorter wavelength} \implies \text{larger momentum and energy} \implies \text{larger accelerators.} \tag{1.26}$$

This is also where the lecture's practical questions enter. If the accelerating field per meter is limited, then to gain more beam energy we must build a longer accelerator, at least roughly linearly in the case of a linear machine. Very roughly speaking, improving the resolution forces us toward larger machines. One may then ask whether there is any limit to the descent to smaller and smaller distances. The lecture gives a cautious answer: perhaps near the Planck length something qualitatively new happens, but we are still enormously far from that scale.

Cosmic rays do reach very high energies, higher than man-made accelerators, but the lecture explains why they are not a replacement for controlled experiments. A rare ultra-high-energy particle hitting a stationary target is not the same as a controlled high-flux collider. To do good experiments one needs enough particles, enough collisions, and enough control to learn something systematic.

So the conclusion is not merely that short wavelength is useful. It is that the whole historical progress of particle physics has been governed by the drive toward shorter wavelength, higher momentum, higher energy, and smaller structure.

1.7 Summary

The lecture begins with the ancient opposition between discrete particles and continuous fields, and it refuses to throw either picture away too quickly. Dalton's chemistry provides the first real evidence that matter comes in discrete units. Radioactivity then breaks the complacent picture of

the elements as ultimate building blocks and produces the first particle beam, already separated into α, β, and γ components.

Gamma radiation leads us to light, but light is first rebuilt as a classical wave. Only after interference and Maxwell's electromagnetic picture are secure does the lecture introduce the wave variables λ, T, f, and ω. Those variables are then carried into quantum mechanics, where photons arrive one at a time but are distributed according to wave-like probabilities, and where each photon carries energy $E_\gamma = \hbar\omega$.

Relativity then supplies the second half of the subject. The modern meaning of $E = mc^2$ is clarified, natural units remove arbitrary conversion factors, and the final derivation

$$p_\gamma = \frac{h}{\lambda} \tag{1.27}$$

closes the loop. Once that equation is in hand, the operational logic of particle physics is unavoidable: to probe smaller objects we need shorter wavelengths, and shorter wavelengths demand larger momentum and larger energy. That, in the lecture's sense, is the pattern that has governed particle physics for roughly a century.

Chapter 2

Waves, Quanta, and the First Picture of a Quantum Field

Quantum field theory is the real tool of the trade, but the lecture begins by postponing its definition. We first put a small stock of mathematics on the board, then use it to connect waves to photons, then push the same logic toward matter waves, periodic space, and finally the harmonic oscillator language out of which the first elementary picture of a quantum field is built. The argument is deliberately staged, and these notes keep that order, with curation by LazyingArt LLC, so that each formula arrives for the same reason it arrives in the lecture.

2.1 Waves as the First Field Language

The lecture begins with waves because waves are already fields. A field is something spread through space, something whose value can vary from point to point. A wave is then a particular field configuration, one that oscillates as it moves. Before we can say what a quantum field is, we need the simplest mathematics of oscillation.

Let us begin with the defining property of an exponential. If

$$f(x) = e^{\alpha x},$$

then the rate of change of f is proportional to f itself:

$$\frac{d}{dx} e^{\alpha x} = \alpha e^{\alpha x}. \tag{2.1}$$

That is the distinctive feature. Over each small interval in x, the function changes by an amount proportional to what is already there.

Sines and cosines behave differently. They oscillate, and under differentiation they rotate into one another:

$$\frac{d}{dx} \sin(kx) = k \cos(kx), \tag{2.2}$$

$$\frac{d}{dx} \cos(kx) = -k \sin(kx). \tag{2.3}$$

Neither one is an exponential by itself. But together they build one. If we differentiate the combination $\cos(kx) + i\sin(kx)$, we find

$$\frac{d}{dx}\big(\cos(kx) + i\sin(kx)\big) = -k\sin(kx) + ik\cos(kx) \tag{2.4}$$

$$= ik\big(\cos(kx) + i\sin(kx)\big). \tag{2.5}$$

So this combination has exactly the exponential property, and we identify it as

$$e^{ikx} = \cos(kx) + i\sin(kx). \tag{2.6}$$

This is not decoration. It is the compact language in which oscillation in space and oscillation in time will be written from here on. If x is a spatial coordinate, we are looking at a spatial oscillation. If x is replaced by time, we are looking at a temporal oscillation. The later formulas for wavelength, frequency, and even the disguised form of the Schrödinger equation all rest on this first bit of blackboard review.

2.2 Frequency, Wavelength, and the Momentum of a Photon

Once the oscillatory language is in place, the lecture turns to the ordinary bookkeeping of waves. The frequency f counts cycles per second. The angular frequency ω measures the same oscillation in radians per second:

$$\omega = 2\pi f. \tag{2.7}$$

The wavelength is denoted by λ. The board writing for the word itself is slightly compressed in the frame, but the intended definition is clear from the lecture: λ is the wavelength.

For a light wave there is one more relation, and now we are no longer speaking of a mere definition. A light wave moves at the speed c, so the distance traveled in one cycle multiplied by the number of cycles per second gives the velocity:

$$\lambda f = c, \tag{2.8}$$

$$\frac{f}{c} = \frac{1}{\lambda}. \tag{2.9}$$

Quantum mechanics now enters through Planck's constant. The lecture uses both h and $\hbar$, switching between them to keep factors of 2π under control:

$$\hbar = \frac{h}{2\pi}. \tag{2.10}$$

What matters is the statement that one quantum of a wave of frequency f, or angular frequency ω, has energy

$$E = \hbar\omega = hf. \tag{2.11}$$

This is not the energy of an entire classical electromagnetic wave. It is the energy of one photon.

Now the lecture pivots to momentum. For a slowly moving particle we write

$$\vec{p} = m\vec{v}, \tag{2.12}$$

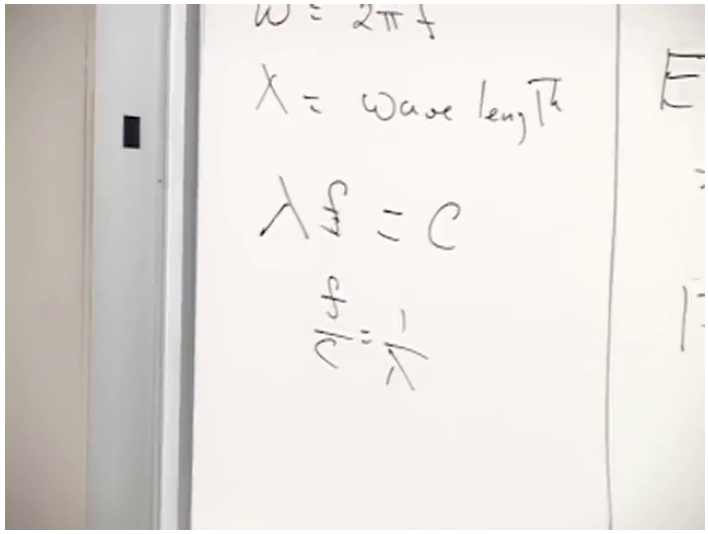

Figure 2.1: Wave relations used in the photon-momentum step.

but this cannot be the correct formula for a photon, because a photon has no rest mass. For a collimated beam of electromagnetic radiation, and therefore for one photon in that beam, the magnitude of the momentum is instead

$$|p| = \frac{E}{c}. \tag{2.13}$$

Substituting the quantum energy and then the wave relation gives

$$|p_{\text{photon}}| = \frac{E}{c} = \frac{hf}{c} = h\frac{f}{c} = \frac{h}{\lambda}. \tag{2.14}$$

This is the first real payoff of the lecture. The formula itself is compact, but the moral is the real point: short wavelength means large momentum. If we want to probe small distances, we must pay dynamically for it. That is one of the central facts behind high-energy particle physics.

2.2.1 Question & Answer

Where does $E = \hbar\omega$ come from? The lecture does not prove that relation here. Susskind answers directly that it comes from the quantum mechanics of the harmonic oscillator. We will return to the harmonic oscillator later in the lecture, and when we do, this formula will no longer look accidental. For the moment we take it as a fact that the rest of the discussion will use.

2.3 Amplitude in Wave Language and Photon Language

At this point the lecture is interrupted by exactly the right question. We have talked about wavelength, frequency, energy, and momentum, but a wave has another obvious feature: amplitude. If we compare $\sin(kx)$ with $A\sin(kx)$, the wavelength is unchanged and the frequency is unchanged; what has changed is how large the oscillation is.

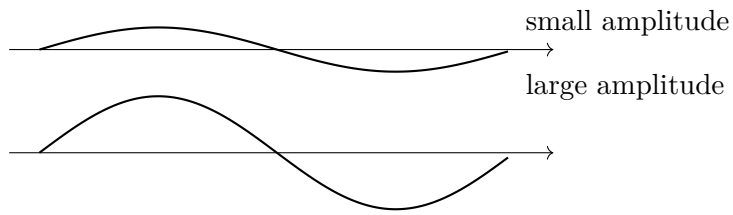

Figure 2.2: Two waves with the same frequency and wavelength but different amplitudes.

2.3.1 Question & Answer

What does the amplitude of a wave mean once we speak in photons? The lecture answers this first at the classical level. For a wave, and in particular for an electromagnetic wave, the energy is proportional to the square of the field strength, hence to the square of the amplitude:

$$E_{\text{wave}} \propto A^2. \tag{2.15}$$

Quantum mechanically, if the wave contains n photons of one frequency ω, then its energy is also

$$E_{\text{wave}} = n\hbar\omega. \tag{2.16}$$

Combining the two statements, we get the lecture's order-of-magnitude rule:

$$A^2 \propto n\hbar\omega. \tag{2.17}$$

In particular, at fixed frequency,

$$A \propto \sqrt{n}. \tag{2.18}$$

This must be kept in the cautious form used in the lecture. The exact normalization is not worked out here. One also has to understand the proportionality as referring to a fixed region of the wave, for example one wavelength, because otherwise the total energy can be made arbitrarily large simply by taking a longer stretch of wave. The audience explicitly raises this point, and the answer is that the lecture is indeed speaking only proportionally, not claiming an exact global formula.

There is another subtlety. In quantum mechanics the question is not perfectly sharp, because amplitude and phase are not simultaneously exact observables. So for one photon one should not speak as though there were a precisely defined classical amplitude attached to it. The safest lecture-faithful statement is the weaker one: for $n = 1$, the same formula gives the order of magnitude.

The lecture also pauses to clear up a common visual confusion. In the usual sketch of an electromagnetic wave moving along the x-axis, the vertical direction in the sine-wave drawing is not another spatial direction in which the photon wanders. It is the value of the electric or magnetic field. Only one axis in that picture is the direction of propagation.

2.4 Which Relations Are General, and Which Are Special to Light?

Having used the photon example, the lecture slows down and sorts the formulas on the board. This is an important beat in the argument. If we do not stop here, the later transition from photons to electrons will look like a jump instead of a controlled generalization.

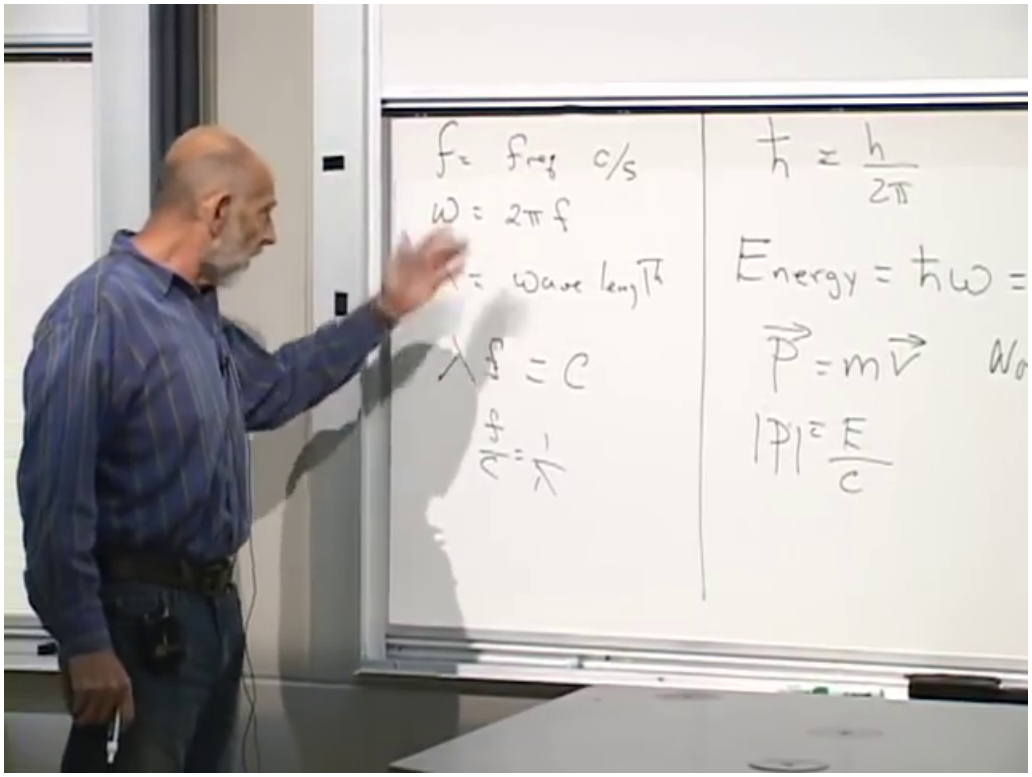

Figure 2.3: Board layout separating wave definitions on the left from quantum and momentum formulas on the right.

The board itself makes the point visible. On the left sit the definitional wave quantities: f, ω, and λ. On the right sit the quantum and momentum relations. But not everything on the board has the same scope.

The definitions remain definitions. The relation

$$\omega = 2\pi f \tag{2.19}$$

is always true because it only translates one frequency unit into another. Likewise λ is just the wavelength by definition.

But the relations containing an explicit c are being fenced off. The equation $\lambda f = c$ is special to waves moving at the speed of light. The momentum formula $|p| = E/c$ is likewise special to radiation moving at the speed of light. On the other hand, the quantum relation $E = \hbar\omega$ is much more general, and the lecture insists that the really lasting momentum statement is

$$p = \frac{h}{\lambda}. \tag{2.20}$$

This is exactly why the lecture lingers over the formula

$$p_{\text{photon}} = \frac{hf}{c} = \frac{h}{\lambda}. \tag{2.21}$$

The first expression still remembers that we are talking about a photon. The last one is the broader de Broglie relation. The same momentum-wavelength connection will be carried forward to slowly moving matter, whereas the explicit factor of c will not.

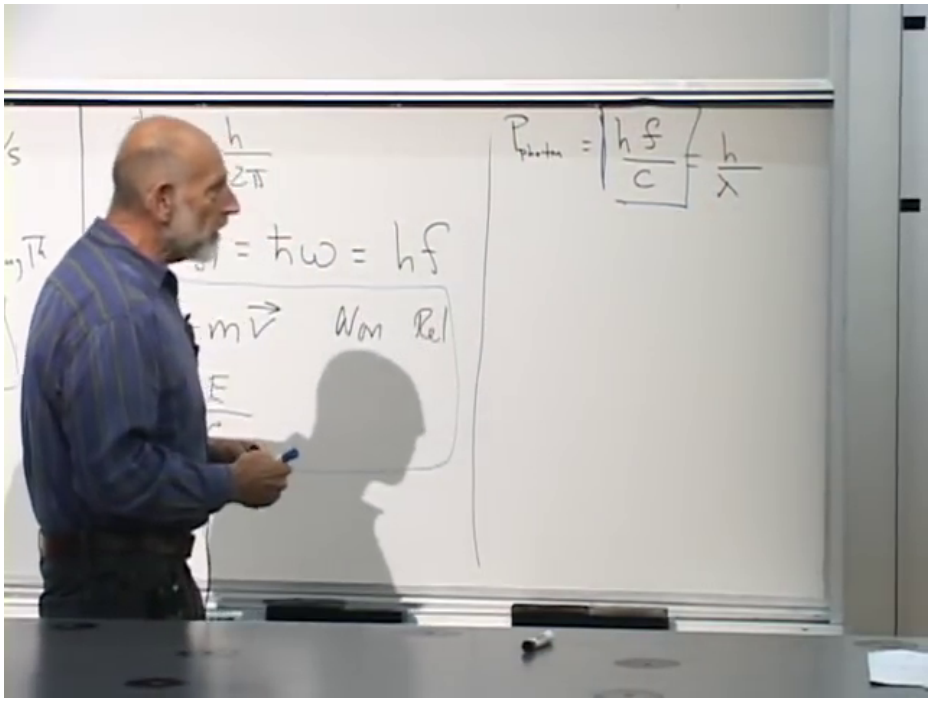

Figure 2.4: The moment at which the photon result is rewritten as the more general wavelength formula.

The lecture also remarks that $\vec{p} = m\vec{v}$ is itself only a slow-particle approximation. There is an interpolation between that formula and the lightlike one, but it is not part of tonight's story. Tonight the contrast is between the nonrelativistic regime and the wave moving at the speed of light.

2.5 Slow Matter Waves: Frequency, Wavelength, and Velocity

Once $p = h/\lambda$ has been promoted to a general statement, the next question is unavoidable. If an electron is also described by a wave, and if its wavelength is still tied to momentum by $p = h/\lambda$, then what replaces $\lambda f = c$ for a slowly moving electron, nucleus, or even a bowling ball?

Let us write the electron wave as ψ. For a slowly moving particle, the needed extra fact is not a new wave formula but the nonrelativistic relation between energy and momentum:

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}. \tag{2.22}$$

Now combine this with the wave-particle relations

$$E = hf, \qquad p = \frac{h}{\lambda}. \tag{2.23}$$

Then

$$hf = \frac{p^2}{2m} = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2, \tag{2.24}$$

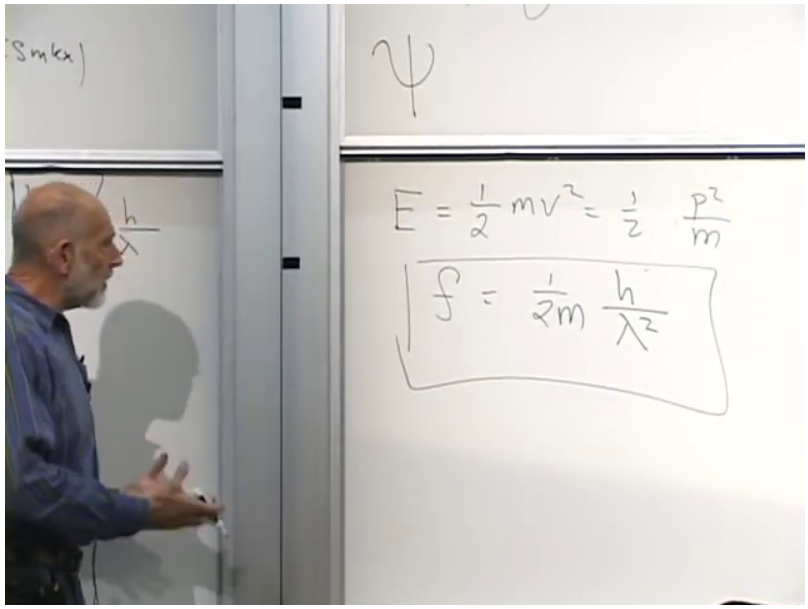

Figure 2.5: Board layout for the nonrelativistic energy formula and the resulting matter-wave frequency relation.

so after dividing by h,

$$f = \frac{h}{2m\lambda^2}. \tag{2.25}$$

This is the new frequency-wavelength relation for a slowly moving particle. It is strikingly different from the light-wave formula. For light,

$$f \propto \frac{1}{\lambda}.$$

For a nonrelativistic matter wave,

$$f \propto \frac{1}{\lambda^2}.$$

The lecture explicitly warns us not to memorize the result in isolation. Its importance is that it can always be rederived from two ingredients: the appropriate energy-momentum relation and the general de Broglie relation $p = h/\lambda$.

2.5.1 Question & Answer

Which velocity of a matter wave packet should be identified with the particle? Now the lecture pauses again, because a natural but misleading question arises. If we know f and λ, can we not simply multiply them and get the velocity? We can, but not the velocity of the particle itself.

For one monochromatic component, the phase velocity is

$$v_\phi = f\lambda = \frac{h}{2m\lambda}. \tag{2.26}$$

Because this depends on λ, different wavelength components travel with different phase velocities. That is the hallmark of a dispersive medium.

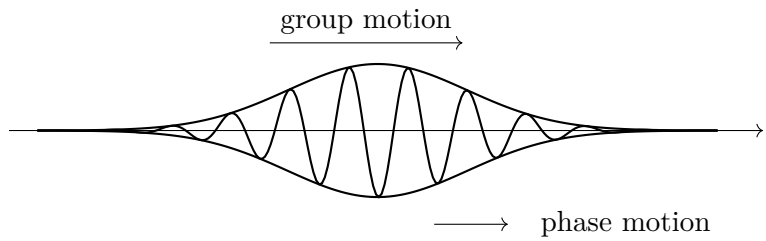

Figure 2.6: A schematic wave packet: the envelope and the interior crests generally move at different velocities.

The lecture therefore distinguishes two velocities. The phase velocity is the motion of the internal crests and troughs. The group velocity is the motion of the packet as a whole. It is the group velocity that we identify with the particle.

This is also the moment at which the lecture remarks that the result is really the Schrödinger equation in compressed form. Frequency encodes the time dependence of the wave; wavelength encodes its spatial dependence. An equation relating the two is already most of what the free Schrödinger equation does.

There is one important special case. If every wavelength moves with the same speed, then the crests and the packet move together. That is what happens for light in vacuum. There phase and group velocity coincide. For the nonrelativistic matter-wave relation above, they do not. The lecture even notes that the phase velocity may exceed the speed of light, but that carries no information; the physically relevant signal velocity is the group velocity.

2.6 Finite Space, Periodic Space, and Quantized Momentum

The lecture now changes scale and motivation. Up to this point we have been talking as though space were infinite, but the next step insists on a standard piece of good practice: first make the system finite, then understand the limit. Susskind's practical slogan is memorable and exact in spirit. Computers do not like infinity.

So let us reduce the problem to one dimension. There is a tempting first way to make space finite: take a segment of length L, fix the ends, and let the wave bounce back and forth like a violin string. This is perfectly respectable physics, but it comes with a drawback. A wave moving to the right carries momentum to the right. When it reflects, the momentum flips sign. So the boundary has destroyed momentum conservation.

2.6.1 Question & Answer

How do we make space finite without destroying momentum conservation? The lecture's answer is to identify the two ends. One may picture this as a circle if one likes, but only schematically. The important point is not ordinary geometry; it is periodicity. When the wave reaches one end, it reappears at the other and continues in the same direction. The endpoint is no longer special.

Once space is periodic, the allowed wavelengths are discrete. An integer number of wavelengths

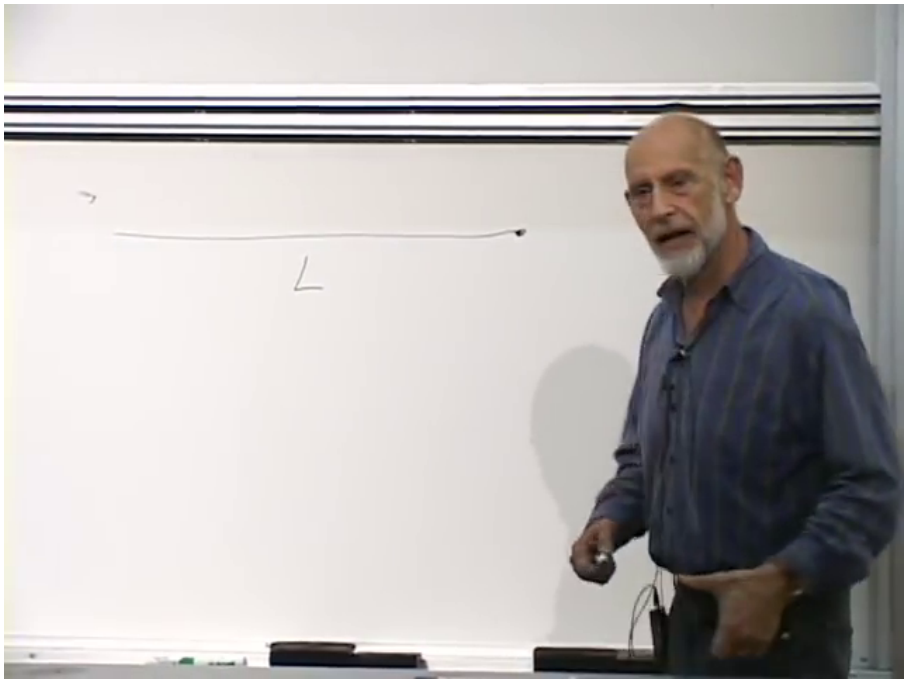

Figure 2.7: The board sketch with total length L, presented as a schematic path rather than a literal geometric circle.

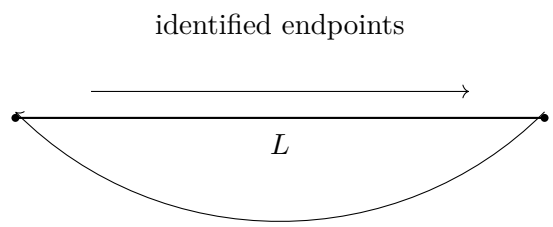

Figure 2.8: A cleaned reconstruction of the periodic identification. The picture is schematic; the point is periodicity, not an embedded circle in two-dimensional space.

must fit around the total length:

$$\lambda_N = \frac{L}{N}, \qquad N = 1, 2, 3, \ldots \tag{2.27}$$

and therefore the allowed momenta are also discrete:

$$p_N = \frac{h}{\lambda_N} = \frac{Nh}{L}. \tag{2.28}$$

This is the lecture's first clean example of momentum quantization.

If L is very large, the spacing h/L between allowed momenta is very small, so the spectrum looks almost continuous. If L is made smaller, the spacing grows. But the main point is structural: momentum comes in discrete multiples because only certain wavelengths are compatible with periodicity.

The audience then asks about a packetized wave, and the answer is equally important. A wave packet is a superposition of plane waves. In periodic space, that means a superposition of the allowed discrete modes. This is simply Fourier analysis with a discrete momentum basis.

The lecture also recasts the same fact for a genuine circular wire of radius r, with circumference $L = 2\pi r$. To avoid the notational stumble that occurs on the board, let us keep N for the mode label and J for angular momentum. Then

$$p_N = \frac{Nh}{2\pi r}, \qquad J = rp_N = N\hbar. \tag{2.29}$$

So angular momentum is quantized in units of $\hbar$, while the momentum spacing itself depends on the size of the circle.

2.7 Quantum Mechanics, the Harmonic Oscillator, and the First Picture of a Field

After the break the lecture says, in effect, that we now have to enter the real world of quantum mechanics a little bit. The first object introduced is the commutator:

$$[A, B] = AB - BA. \tag{2.30}$$

Its reverse-order version differs by a sign,

$$[B, A] = -[A, B]. \tag{2.31}$$

Here the commutator is an algebraic operation, not a temporal instruction to measure one thing and then another. The lecture uses it as the test for simultaneous measurability: if two observables commute, they can be measured together; if they do not, the attempt to measure one interferes with the other.

For an ordinary nonrelativistic particle, the position components commute with one another, and the momentum components commute with one another:

$$[x, y] = [y, z] = [z, x] = 0, \qquad [p_x, p_y] = [p_y, p_z] = [p_z, p_x] = 0. \tag{2.32}$$

But the corresponding position and momentum components do not:

$$[x, p_x] = [y, p_y] = [z, p_z] = i\hbar. \tag{2.33}$$

Mixed components such as x and p_y do commute. The lecture briefly notes that this simple position-momentum discussion applies most cleanly to nonrelativistic particles; for photons the notion of position is subtler, and that subtlety is postponed.

The real destination is the harmonic oscillator. It appears here because waves oscillate. A field mode of definite frequency is already halfway to an oscillator. The quantum-mechanical novelty is that the oscillator does not carry arbitrary energy. Its energy levels are discrete. In the lecture the zero of energy is shifted so that the lowest level sits at 0, not at $\frac{1}{2}\hbar\omega$. With that convention,

$$E_n = n\hbar\omega, \qquad n = 0, 1, 2, \ldots \tag{2.34}$$

and the state with energy E_n is denoted by the ket $|n\rangle$.

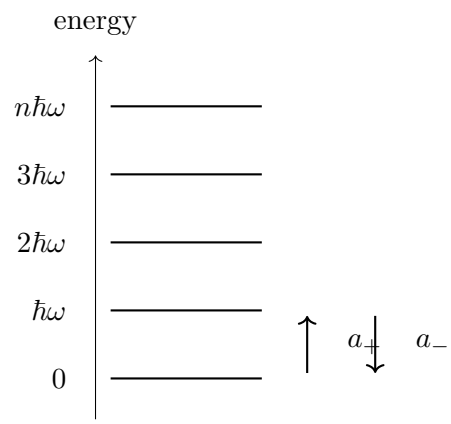

Figure 2.9: A cleaned energy-level picture of the shifted harmonic oscillator used in the lecture.

Now come the raising and lowering operators, written on the board as $A+$ and $A-$, and here written as a_+ and a_-:

$$a_+|n\rangle = \sqrt{n+1}\,|n+1\rangle, \tag{2.35}$$
$$a_-|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{2.36}$$
$$a_-|0\rangle = 0. \tag{2.37}$$

These operators move us one step up or down the ladder of states, thereby adding or removing one quantum of energy of frequency ω.

Their algebra is already revealing. Acting in one order gives

$$a_+a_-|n\rangle = n|n\rangle, \tag{2.38}$$

while the reverse order gives

$$a_-a_+|n\rangle = (n+1)|n\rangle. \tag{2.39}$$

So the two orders do not agree. Applied to every state, this implies

$$[a_-, a_+] = 1. \tag{2.40}$$

In the shifted convention being used here, the Hamiltonian may be written as

$$H = \hbar\omega\, a_+a_-. \tag{2.41}$$

What do these operators mean physically? The lecture is careful here. One does not ordinarily measure a_+ and a_- separately as direct observables. Their noncommutativity reflects the fact that they encode amplitude-and-phase information that cannot be made simultaneously sharp. But mathematically they do something definite: they create and annihilate quanta. If the oscillator is an electromagnetic mode, the quanta are photons.

Now the lecture returns to the periodic world of length L. Each allowed wavelength $\lambda_N = L/N$ carries an allowed frequency ω_N. Each such mode is one harmonic oscillator. A field is therefore a whole collection of oscillators, one for each allowed mode. The state of the field is not labeled by one integer but by an infinite list:

$$|n_1, n_2, n_3, \ldots\rangle. \tag{2.42}$$

Here the notation has to be kept clean. The mode label is N. The occupation number of a given mode is n_N. The audience notices the danger of confusing those two integers, and the lecture explicitly pauses to separate them.

The violin-string analogy is the right one. There is a fundamental mode, then higher harmonics, then higher still. Each mode has its own frequency, and each mode can carry zero, one, two, or more quanta. The total energy of the field is then built mode by mode:

$$E = \sum_N n_N \hbar \omega_N. \tag{2.43}$$

A quantum field, in this first elementary picture, is exactly this: a collection of harmonic oscillators together with creation and annihilation operators for each oscillator.

That is why the lecture ends by tying the whole discussion back to particle physics experiments. Particles with given energy and momentum come in. Particles with other energies and momenta go out. Mathematically, we remove quanta from an initial state and add quanta to a final state. Creation and annihilation operators are the symbols that do that work. They are the language in which particle processes are written.

2.8 Summary

The lecture opens by delaying the definition of quantum field theory and instead assembling the mathematics of oscillation that the subject keeps using. From exponentials, sines, and cosines we move to frequency and wavelength, from there to the one-quantum energy $E = \hbar\omega$, and then to the photon momentum formula $p = h/\lambda$. After the audience's question about amplitude, the lecture sorts the formulas by generality and then turns the same wave-particle dictionary onto slowly moving matter, where the relation $f \propto 1/\lambda^2$ leads naturally to the distinction between phase velocity and group velocity. Periodic space then makes momentum discrete without sacrificing momentum conservation. After the break, the harmonic oscillator provides the missing quantum-mechanical machinery, and the promised picture finally appears: a quantum field is a collection of oscillators, one for each allowed mode, together with operators that add and remove quanta in those modes.

Chapter 3

What a Quantum Field Is

This chapter follows Leonard Susskind's third lecture in *Particle Physics 1: Basic Concepts*, curated by LazyingArt LLC. We begin exactly where the lecture begins: last time we were beginning to get to the idea of a quantum field, and now we want to answer two linked questions. What is a quantum field, and how is it related to particles? The lecture does not answer that by postulate. It rebuilds the answer from periodic space, harmonic oscillators, occupation numbers, and Fourier modes, and only then allows the field itself to emerge.

3.1 Periodic Space, Allowed Wave Numbers, and Momentum

We begin, not with operators, but with periodic space. The circle is a trick, but it is a very useful trick. If we put a system in a finite interval with hard walls, waves reflect, and momentum is no longer the right organizing principle. If instead we identify the ends of the interval, a wave can move around the circle without reflecting. That is not merely an analog of momentum conservation; for this finite box it is the way to preserve momentum conservation while avoiding the complications of infinite volume.

A plane wave on a circle of circumference L has the form e^{ikx}. Periodicity requires that after traveling once around the circle the wave come back to its original value:

$$e^{ikL} = 1. \tag{3.1}$$

Since $e^{i2\pi n} = 1$ for every integer n, the allowed wave numbers satisfy

$$kL = 2\pi n, \qquad k = \frac{2\pi n}{L}, \qquad n \in \mathbb{Z}. \tag{3.2}$$

The quantity k is the wave number. It is important to keep track of what is and is not an integer here. The integer is n. The wave number k itself is an integer multiple of $2\pi/L$, not an integer.

The same condition can be stated in terms of wavelength. If λ is the distance for one full cycle, then one full cycle means $k\lambda = 2\pi$, so

$$k\lambda = 2\pi, \qquad k = \frac{2\pi}{\lambda}. \tag{3.3}$$

Thus large wavelength means small wave number, and small wavelength means large wave number. The lecture then reconnects this to ordinary quantum mechanics: for a quantum-mechanical wave,

momentum and wave number are related by

$$p = \hbar k. \tag{3.4}$$

In units with $\hbar = 1$, one often speaks loosely of k as the momentum. That is the lecture's practice too, but only after the distinction has been made clearly.

We should also keep one more point in view, because the late questions return to it. Making space periodic does not discretize position. What becomes discrete is the set of allowed momenta. The positions remain points on the circle. Later, when L is made larger and larger, the spacing $2\pi/L$ between neighboring allowed values of k becomes smaller and smaller, and the discrete sum over modes tends toward an integral.

3.2 Occupation Numbers and Fock-Space State Labels

Once the allowed momenta are in place, the lecture asks the next natural question: how many quanta occupy each allowed momentum mode? Here Susskind slows down, because the notation can become misleading if we are careless. One integer, n, labels the allowed values $k = 2\pi n/L$. A different integer tells us how many quanta sit in a given mode. The whole transition to field language depends on not confusing these two roles.

The second integer is the occupation number, written $n(k)$. It means the number of particles, or quanta, with wave number k. A many-particle state is therefore specified by giving the full list of occupation numbers:

$$\big|\{n(k)\}\big\rangle = \big|\,\ldots, n(-2), n(-1), n(0), n(1), n(2), \ldots\big\rangle. \tag{3.5}$$

The lecture first writes this schematically as a string of occupations for the first, second, third, and so on allowed momenta, and then reminds us that the string must extend in both directions, because both positive and negative momenta occur.

The vacuum is simply the state in which every slot is empty:

$$|0\rangle = |\,\ldots, 0, 0, 0, 0, 0, \ldots\rangle. \tag{3.6}$$

At this stage the lecture is very concrete. Nothing has yet been said about interactions, or even about the field itself. We only have a set of allowed momentum modes and a way to specify how many quanta occupy each of them. But this is already a genuine many-particle state space, and it is the right one for quantum fields.

3.3 Mode Operators $a^\dagger(k)$ and $a(k)$

Now that the states have been labeled, the next step is to introduce operators that act on those labels. On the board Susskind writes $a^+(k)$ and $a^-(k)$. We will use the standard notation $a^\dagger(k)$ and $a(k)$, but the meaning is the same: for each momentum mode k, there is one operator that raises the occupation number and one that lowers it.

Their action on an occupation-number state is

$$a^\dagger(k)\big|\,\ldots, n(k), \ldots\big\rangle = \sqrt{n(k)+1}\,\big|\,\ldots, n(k)+1, \ldots\big\rangle, \tag{3.7}$$

$$a(k)\big|\,\ldots, n(k), \ldots\big\rangle = \sqrt{n(k)}\,\big|\,\ldots, n(k)-1, \ldots\big\rangle. \tag{3.8}$$

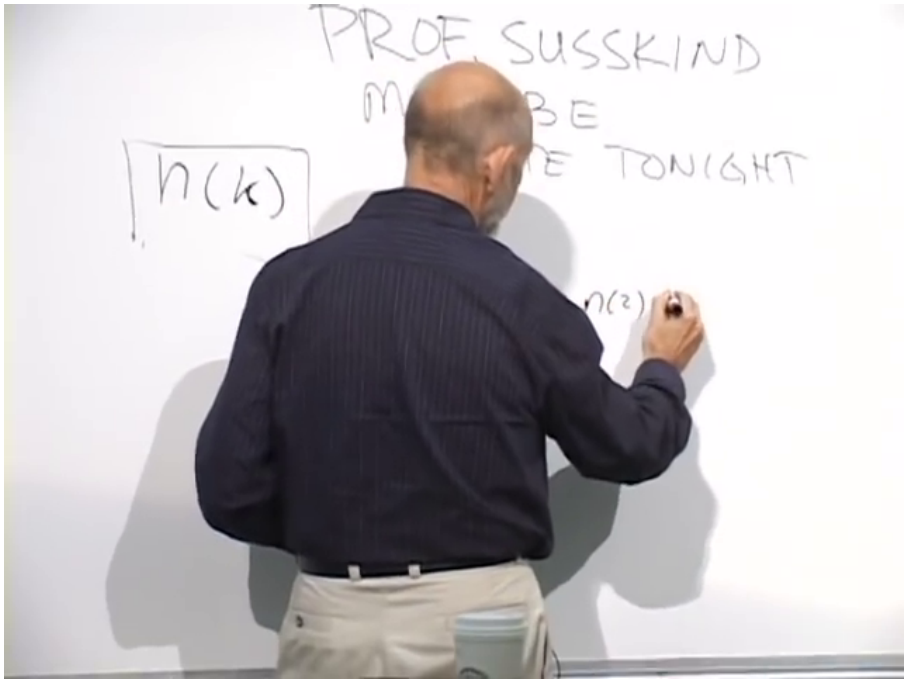

Figure 3.1: Occupation-number notation $n(k)$ on the board. The board shows the bookkeeping device; the text writes the state cleanly as a full list of occupations.

The lecture briefly stumbles over the wording and then corrects itself in exactly the right way: $a^\dagger(k)$ goes to the k-th slot and increases that occupation number by one unit, while $a(k)$ decreases it, bringing in the appropriate square root.

A one-particle momentum eigenstate is obtained by acting on the vacuum with a single creation operator:

$$|k\rangle = a^\dagger(k)|0\rangle. \tag{3.9}$$

With this notation, the lecture immediately shows how the operator language can describe scattering. If a particle initially in mode k_7 is scattered into mode k_9, then

$$a^\dagger(k_9)a(k_7)|k_7\rangle = |k_9\rangle. \tag{3.10}$$

Why do the square roots not show up here? Because $a(k_7)$ acts on a slot whose occupation number is 1, giving $\sqrt{1} = 1$, and $a^\dagger(k_9)$ creates into an empty slot, again giving $\sqrt{0+1} = 1$.

The same language can also describe processes in which particle number changes. That is the point at which single-particle quantum mechanics really runs out of room. Schematically,

$$a^\dagger(k_9)a^\dagger(k_{16})a(k_7)|k_7\rangle = |k_9, k_{16}\rangle. \tag{3.11}$$

One particle goes in, two come out. The lecture is careful to say that this is not yet a law of nature; it is a mathematical description of a possible kind of process. If the dynamics allows particle number to change, this is the language in which such changes are naturally expressed.

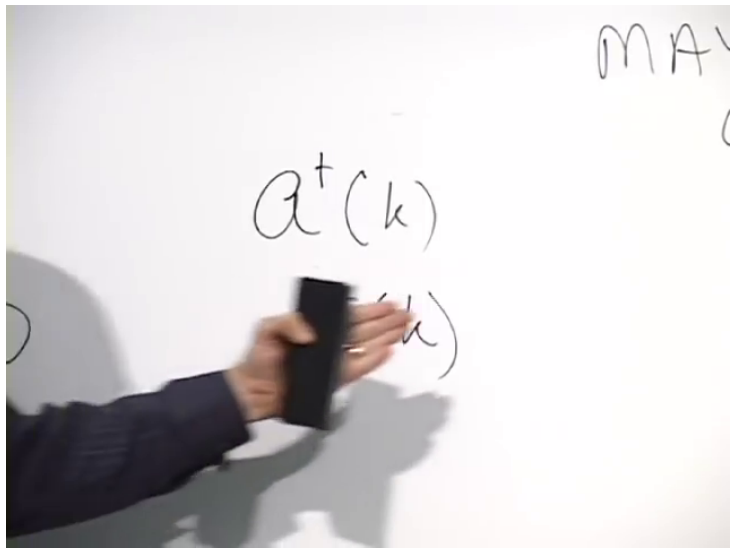

Figure 3.2: Mode-by-mode creation and annihilation operators on the board. The lecture writes $a^+(k)$ and $a^-(k)$; the notes use $a^\dagger(k)$ and $a(k)$.

3.4 From Classical Fourier Coefficients to the Quantum Field

Only now does the lecture ask the decisive question: what is the connection between the field and the a's? The answer is given by moving briefly back to the classical picture. A classical field on a periodic interval can be expanded in Fourier modes,

$$\psi(x) = \sum_k \alpha(k) e^{ikx}, \tag{3.12}$$

$$\psi^*(x) = \sum_k \alpha^*(k) e^{-ikx}. \tag{3.13}$$

The coefficients $\alpha(k)$ are just the Fourier amplitudes of the classical wave. Since the real and imaginary parts of those amplitudes can be measured, the lecture treats them as classical observables.

The quantum step is then stated quite directly: the a's are the quantum-mechanical versions of the Fourier coefficients. The lecture does not write an ordinary equality sign here, because these are not the same kinds of objects. Instead, it treats quantization as replacement:

$$\alpha(k) \rightsquigarrow a(k), \qquad \alpha^*(k) \rightsquigarrow a^\dagger(k). \tag{3.14}$$

Susskind openly hesitates for a moment about which convention is standard and then flips to the standard choice. That hesitation is pedagogically useful. It reminds us that the sign conventions are conventions. In the rest of the chapter we fix

$$\Psi(x) = \sum_k a(k) e^{ikx}, \tag{3.15}$$

$$\Psi^\dagger(x) = \sum_k a^\dagger(k) e^{-ikx}. \tag{3.16}$$

This is the point at which the field finally appears: a quantum field is an operator-valued Fourier expansion.

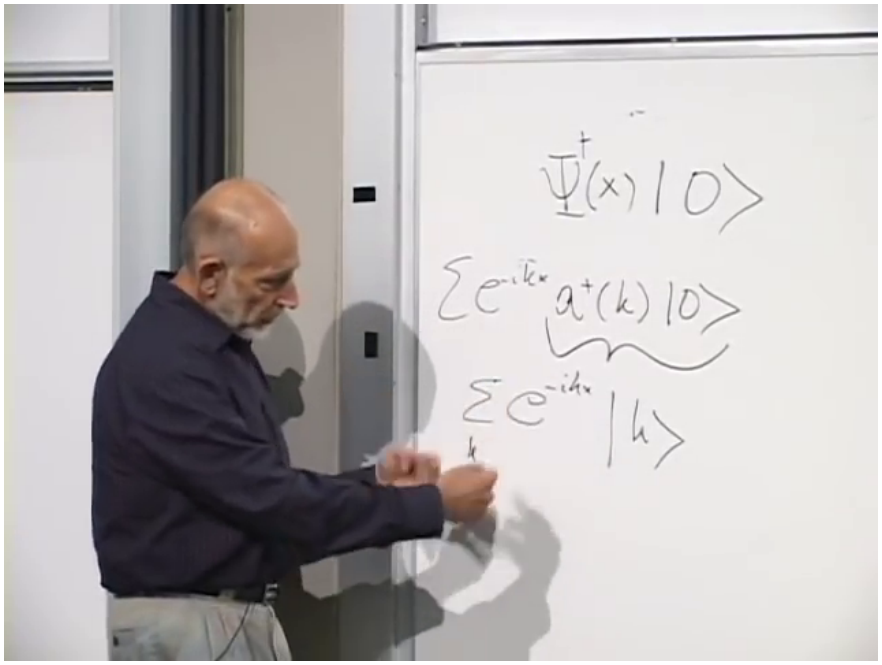

Figure 3.3: $\Psi^\dagger(x)$ acting on the vacuum. The board first expands the field in momentum modes and then rewrites the result directly in the one-particle momentum basis.

There is also an important warning at this stage. Creation and annihilation operators are mathematical operators. Their existence does not mean that the laws of nature must create and destroy particles. Whether particle number changes is a dynamical question. But if the laws of nature do permit such processes, then these operators are the right tools for describing them. The lecture refuses to let the terminology do more work than the mathematics has earned.

3.4.1 Question & Answer

What does $\Psi^\dagger(x)$ actually create?

This is the cleanest conceptual obstacle in the lecture, and it is resolved by a direct calculation. First, the annihilation field kills the vacuum:

$$\Psi(x)|0\rangle = 0. \tag{3.17}$$

There is no particle present to annihilate. The creation field is more interesting. Acting on the vacuum, it gives

$$\Psi^\dagger(x)|0\rangle = \sum_k e^{-ikx} a^\dagger(k)|0\rangle \tag{3.18}$$

$$= \sum_k e^{-ikx}|k\rangle, \tag{3.19}$$

where

$$|k\rangle = a^\dagger(k)|0\rangle. \tag{3.20}$$

At first this looks like a strange object: a sum of momentum eigenstates with phases e^{-ikx}. But that is exactly the construction of a position state in one-particle quantum mechanics. If we superpose

momentum states with those phases, we get a state localized at position x. So the lecture answers its own question in the natural way:

$$\Psi^\dagger(x) \text{ creates a particle at position } x. \tag{3.21}$$

Equivalently, $\Psi(x)$ annihilates a particle at position x.

There is a secondary observation here which the lecture also makes. Since each coefficient e^{-ikx} has unit magnitude, all allowed momenta appear with equal probability weight in this state. So the position state is a coherent superposition spread uniformly over momentum space. If we enlarge the circle and let $L \to \infty$, the sum over k will become an integral, but the interpretation does not change.

3.5 Local Reactions, Bose Enhancement, and Stimulated Emission

Once $\Psi(x)$ and $\Psi^\dagger(x)$ have acquired a position-space meaning, the lecture immediately turns to locality. This is where the tone becomes particularly revealing. Susskind first proposes a ridiculous process on purpose: a particle is absorbed at a sharply localized point x, and then two particles are emitted somewhere far away, one on Alpha Centauri and one on Betelgeuse. That, he says, is not the sort of process we want. It is manifestly non-local.

So what is the right local process? If one particle is absorbed at a point x and two are emitted, then the emission should occur at the same point. That is the motivation for the operator product

$$\Psi^\dagger(x)\Psi^\dagger(x)\Psi(x). \tag{3.22}$$

The first field annihilates the incoming particle at x, and the two creation fields emit the outgoing particles from that same point.

$$\Psi(x) = \sum_q a(q)e^{iqx}, \qquad \Psi^\dagger(x) = \sum_\ell a^\dagger(\ell)e^{-i\ell x}. \tag{3.23}$$

These are precisely the formulas carried on the left board in the lecture, rewritten here in a consistent convention.

Let the initial state be a one-particle momentum eigenstate $|k_{\text{initial}}\rangle$. The annihilation field acts first:

$$\Psi(x)|k_{\text{initial}}\rangle = \sum_q e^{iqx}a(q)a^\dagger(k_{\text{initial}})|0\rangle \tag{3.24}$$

$$= e^{ik_{\text{initial}}x}|0\rangle. \tag{3.25}$$

Only the term matching the occupied mode survives. The lecture stresses this point: an annihilation operator has to find something to annihilate.

Now the two creation fields act on the vacuum:

$$\Psi^\dagger(x)\Psi^\dagger(x)\Psi(x)|k_{\text{initial}}\rangle = e^{ik_{\text{initial}}x}\sum_{\ell,m} e^{-i\ell x}e^{-imx}a^\dagger(\ell)a^\dagger(m)|0\rangle \tag{3.26}$$

$$= \sum_{\ell,m} e^{i(k_{\text{initial}}-\ell-m)x}|\ell, m\rangle. \tag{3.27}$$

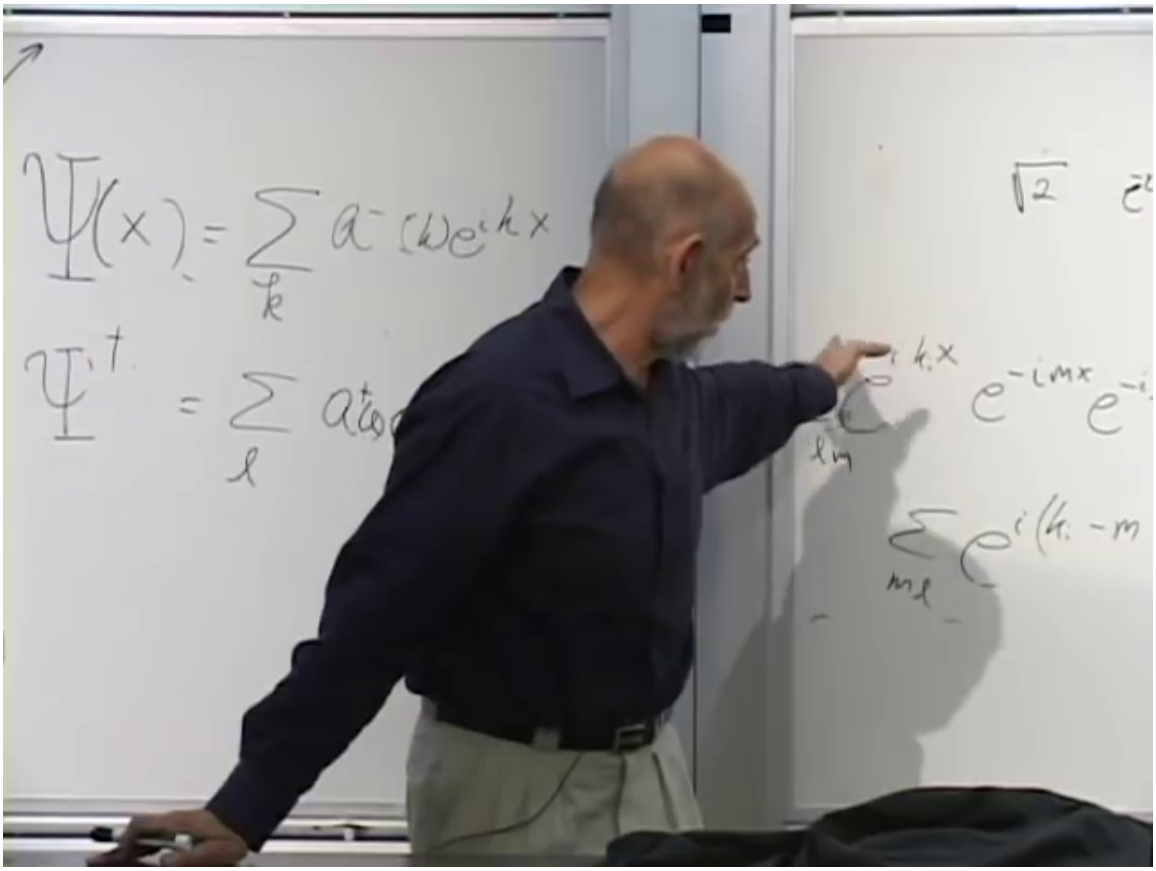

Figure 3.4: The board layout during the locality and probability discussion. On the left are the field expansions; on the right is the momentum-space amplitude for the local $1 \to 2$ process.

This is the basic local amplitude. The final state is a superposition of all two-particle momentum states. In this simplified model we have deliberately allowed the target to absorb recoil, so there is no exact momentum conservation among the particles alone. The phase records the difference between the incoming momentum and the two outgoing momenta, which is exactly the momentum carried away by the target.

At first the lecture draws the conclusion that all final momentum pairs appear with equal probability, since the coefficient is only a phase. Then it pauses and corrects itself. The special case $\ell = m$ is different. If the first creation operator has already created one particle in mode m, then the second creation into that same mode produces an extra factor:

$$a^\dagger(m)|\dots, n(m) = 1, \dots\rangle = \sqrt{2}\,|\dots, n(m) = 2, \dots\rangle. \tag{3.28}$$

So there is a Bose enhancement when the two particles come out in the same state.

This is not the end of the story, however. A student points out that when $\ell \neq m$, the raw double sum contains both ordered terms

$$a^\dagger(\ell)a^\dagger(m)|0\rangle \qquad \text{and} \qquad a^\dagger(m)a^\dagger(\ell)|0\rangle, \tag{3.29}$$

and these are equal because creation operators commute. So the lecture backs away from an overly strong comparison between the same-momentum and different-momentum channels. The safe

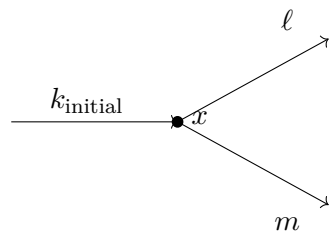

Figure 3.5: Editorial redraw of the local process emphasized in the lecture: one incoming particle is absorbed at the point x, and two outgoing particles are emitted from that same point.

conclusion, and the one worth preserving, is the algebraic one: whenever we create into a mode that is already occupied, an extra square-root factor appears. That is the bosonic effect we should trust.

The lecture then gives the cleaner example. Suppose an atom at the point x emits a photon. If the initial state is the vacuum, then

$$\Psi^\dagger(x)|0\rangle = \sum_k e^{-ikx}|k\rangle, \tag{3.30}$$

so in this simplified model every momentum mode is emitted with equal magnitude. That is the spontaneous process.

Now let there already be one photon in mode ℓ. Then

$$\Psi^\dagger(x)|\ell\rangle = \sum_{k\neq\ell} e^{-ikx}|k,\ell\rangle + e^{-i\ell x}\sqrt{2}\,|2_\ell\rangle, \tag{3.31}$$

where $|2_\ell\rangle$ denotes the state with two quanta in the same mode ℓ. The extra $\sqrt{2}$ is exactly the enhancement seen in the local $1 \to 2$ calculation.

More generally, if the mode ℓ already has occupation number n_ℓ, then

$$a^\dagger(\ell)|\dots, n_\ell, \dots\rangle = \sqrt{n_\ell + 1}\,|\dots, n_\ell + 1, \dots\rangle. \tag{3.32}$$

Therefore the probability for emission into that already occupied mode is enhanced by a factor

$$n_\ell + 1. \tag{3.33}$$

If $n_\ell = 0$, this is spontaneous emission. If n_ℓ is large, the enhancement can be very large. That is stimulated emission.

The lecture is careful not to dress this up in the wrong language. Bosons do not literally attract one another into the same state. The effect is statistical. It comes from the identity of the particles and the square-root factors in the bosonic algebra.

A final technical remark from the lecture belongs here. In more complicated operator products, the order is not always obvious at first sight. But because creation operators commute among themselves and annihilation operators commute among themselves, one can rearrange expressions into normal order, with annihilation operators to the right and creation operators to the left. In that ordering, one first lets the operators eat the incoming particles and then create the outgoing ones.

3.6 Bosonic Algebra, Number Density, and Late Clarifications

Only after these examples does the lecture permit itself to summarize the algebra. This order matters. The commutation relations are not introduced as abstract axioms floating above the physics; they appear after we have already seen what they do.

For bosons,

$$[a^\dagger(k), a^\dagger(\ell)] = 0, \tag{3.34}$$

$$[a(k), a(\ell)] = 0, \tag{3.35}$$

$$[a(k), a^\dagger(\ell)] = \delta_{k\ell}. \tag{3.36}$$

Here $\delta_{k\ell}$ is the Kronecker delta, equal to 1 when $k = \ell$ and 0 otherwise. This is the harmonic-oscillator algebra, repeated for each momentum mode.

The lecture then asks what object is represented by the integral

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x). \tag{3.37}$$

Rather than naming it immediately, it works it out:

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x) = \sum_{k,\ell} a^\dagger(k)a(\ell) \int_0^L dx\, e^{-ikx} e^{i\ell x} \tag{3.38}$$

$$= \sum_{k,\ell} a^\dagger(k)a(\ell) \int_0^L dx\, e^{i(\ell-k)x}. \tag{3.39}$$

Now the periodicity of the circle does the work once again:

$$\int_0^L dx\, e^{i(\ell-k)x} = L\,\delta_{k\ell}. \tag{3.40}$$

So the double sum collapses:

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x) = L \sum_k a^\dagger(k)a(k). \tag{3.41}$$

But $a^\dagger(k)a(k)$ is just the occupation number in the mode k. Hence, with the lecture's normalization convention,

$$\frac{1}{L}\int_0^L dx\, \Psi^\dagger(x)\Psi(x) = \sum_k a^\dagger(k)a(k) = N, \tag{3.42}$$

the total number of particles. The lecture explicitly remarks that other conventions absorb the factor of L into the field. Here we leave the factor visible, because that is how the lecture keeps track of it.

Accordingly, $\Psi^\dagger(x)\Psi(x)$ is the particle-density operator. One might be tempted to call it the number of particles at the point x, but density is the better language.

3.6.1 Question & Answer

Why is the vacuum not the zero vector?

This question is important because the lecture had earlier said that annihilating the vacuum gives zero. That sentence is correct, but only if one distinguishes very carefully between the vacuum state and the null vector.

The vacuum $|0\rangle$ is a genuine state. It is the ground state of the system, the state with no particles in it, and it is normalized:

$$\langle 0|0\rangle = 1. \tag{3.43}$$

The zero vector, by contrast, is not a physical state at all. It is the null element of the vector space.

So when we write

$$a(k)|0\rangle = 0, \tag{3.44}$$

the 0 on the right-hand side is the zero vector. The statement is that annihilating the vacuum produces no state. It is not the statement that the vacuum itself is the zero vector. The lecture insists on this distinction, and then pushes it one step further: the field is not a state either. In quantum field theory the field is an operator, and the states are the things on which it acts.

Several late clarifications now fall into place. In more than one spatial dimension, the periodic box becomes a torus. In two dimensions it is a rectangle with opposite edges identified; in three dimensions it is a cube with opposite faces identified. The point of the construction is still the same as in one dimension: one makes space periodic in order to avoid infinities while keeping momentum as the natural organizing variable. What becomes discrete is momentum, not position.

The lecture also revisits the analogy with the harmonic oscillator. The real and imaginary parts of the field are the analogs of the oscillator's conjugate variables. Up to conventional factors,

$$\Re\Psi \propto \Psi + \Psi^\dagger, \qquad \Im\Psi \propto \frac{\Psi - \Psi^\dagger}{i}. \tag{3.45}$$

These are conjugate quantities, and they do not commute. So one cannot measure all components of the field simultaneously with arbitrary precision. But when the occupation numbers are very large, the amplitudes are large compared with the fixed right-hand side of the commutator. In that regime the field behaves approximately classically. This is why a large number of photons can be described, to very good approximation, by an ordinary classical electromagnetic wave. It is also why an external classical field can be viewed, in the appropriate limit, as a state containing very many quanta.

Finally the lecture previews the fermionic alternative. The key algebraic change is that commutators are replaced by anticommutators:

$$\{A, B\} = AB + BA. \tag{3.46}$$

In the bosonic theory, many particles can pile into the same state and produce Bose enhancement. In the fermionic theory the Pauli exclusion principle forbids that. Schematically, the bosonic relation

$$[a(k), a^\dagger(\ell)] = \delta_{k\ell} \tag{3.47}$$

is replaced by an anticommutator relation of the same general form, and the result is the opposite statistical behavior: no stimulated emission into an already occupied fermionic state, and no classical field limit obtained by piling indefinitely many quanta into one mode.

3.7 Summary

The lecture has built the simplest bosonic quantum field theory in the order in which one would want to understand it. We begin with a field on a circle, quantize the allowed wave numbers, and connect k, λ, and momentum. We then distinguish sharply between the integer that labels an allowed momentum mode and the occupation number $n(k)$ that counts how many quanta sit in that mode.

From there the harmonic-oscillator operators $a^\dagger(k)$ and $a(k)$ act mode by mode on the Fock-space state labels. The major conceptual step is then to reinterpret those operators as quantized Fourier coefficients. That is what turns the classical wave $\psi(x)$ into the operator-valued field $\Psi(x)$. Acting with $\Psi^\dagger(x)$ on the vacuum produces the familiar momentum superposition that one-particle quantum mechanics recognizes as a position state, so the field becomes an operator that creates and annihilates particles at points in space.

Only after that positional meaning is established does the lecture insist on locality, write the local reaction $\Psi^\dagger(x)\Psi^\dagger(x)\Psi(x)$, and let the bosonic square-root factors show their physical content through Bose enhancement and stimulated emission. At the end the algebra is summarized, the density operator $\Psi^\dagger(x)\Psi(x)$ is identified, and the late clarifications return us to the larger picture: the vacuum is not the zero vector, periodic space is a finite-volume device rather than a lattice ontology, large occupation numbers recover classical field behavior, and fermions will require a different algebra altogether.